

Lecture 1: Contraction Mapping Theorem

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1 Contractions and fixed-point iteration

The contraction statement in Lecture 1 concerns an iteration of the form

$$x_{k+1} = f(x_k), \quad k = 0, 1, 2, \dots,$$

where f maps $\mathbb{R}^n$ into itself. The result is often called the *Banach contraction mapping theorem*. It gives existence and uniqueness of a fixed point and, at the same time, proves that the simple iteration above converges to that point from every starting point.

Definition 1 (Contraction). *Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. A map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a contraction with contraction factor L if there is a number $L \in [0, 1)$ such that*

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in \mathbb{R}^n.$$

The important point is the strict inequality $L < 1$. Differentiability of f is not required for the theorem.

2 The theorem

Theorem 1 (Contraction mapping theorem). *Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfies*

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in \mathbb{R}^n$$

for some $L \in [0, 1)$. Starting from an arbitrary $x_0 \in \mathbb{R}^n$, define $x_{k+1} = f(x_k)$. Then the following statements hold.

- (i) *There is a unique point $x^* \in \mathbb{R}^n$ such that $f(x^*) = x^*$.*
- (ii) *The full sequence (x_k) converges to x^* .*
- (iii) *For every $k \geq 0$,*

$$\|x_k - x^*\| \leq \frac{L^k}{1 - L} \|x_1 - x_0\|. \quad (1)$$

Also,

$$\|x_k - x^*\| \leq L^k \|x_0 - x^*\|. \quad (2)$$

- (iv) *For every $k \geq 1$, the computable a posteriori estimate*

$$\|x_k - x^*\| \leq \frac{L}{1 - L} \|x_k - x_{k-1}\| \quad (3)$$

holds.

3 Detailed proof

Proof. We separate the argument into several steps.

Step 1: successive iterates get closer. For $j \geq 1$, the contraction inequality and the definition of the iteration give

$$\begin{aligned}\|x_{j+1} - x_j\| &= \|f(x_j) - f(x_{j-1})\| \\ &\leq L\|x_j - x_{j-1}\|\end{aligned}$$

Applying this inequality repeatedly yields

$$\|x_{j+1} - x_j\| \leq L^j \|x_1 - x_0\|, \quad j \geq 0. \quad (4)$$

The case $j = 0$ is equality, so the formula is valid for all $j \geq 0$.

Step 2: the sequence is Cauchy. Fix integers $m > k$. By inserting the intermediate iterates and using the triangle inequality,

$$\begin{aligned}\|x_m - x_k\| &\leq \sum_{j=k}^{m-1} \|x_{j+1} - x_j\| \\ &\leq \sum_{j=k}^{m-1} L^j \|x_1 - x_0\| \\ &= L^k \frac{1 - L^{m-k}}{1 - L} \|x_1 - x_0\| \\ &\leq \frac{L^k}{1 - L} \|x_1 - x_0\|.\end{aligned}$$

Because $0 \leq L < 1$, the last expression tends to zero as $k \rightarrow \infty$, uniformly in $m > k$. More explicitly, given $\varepsilon > 0$, choose K so large that

$$\frac{L^K}{1 - L} \|x_1 - x_0\| < \varepsilon.$$

Then $m > k \geq K$ implies $\|x_m - x_k\| < \varepsilon$. This is precisely the Cauchy criterion.

Step 3: completeness supplies a limit. The space $\mathbb{R}^n$ is complete under the Euclidean norm. Since all norms on the finite-dimensional space $\mathbb{R}^n$ are equivalent, it is complete under the chosen norm as well. Therefore the Cauchy sequence (x_k) has a limit; write

$$x_k \longrightarrow x^* \in \mathbb{R}^n.$$

This is the one place where completeness is essential: the limit must remain inside the space on which f is defined.

Step 4: the limit is a fixed point. A contraction is Lipschitz continuous. Consequently,

$$\|f(x_k) - f(x^*)\| \leq L\|x_k - x^*\| \longrightarrow 0,$$

so $f(x_k) \rightarrow f(x^*)$. On the other hand, $f(x_k) = x_{k+1}$, and the shifted sequence (x_{k+1}) has the same limit x^* . Uniqueness of limits now gives

$$f(x^*) = x^*.$$

Thus a fixed point exists and every starting point produces iterates converging to it.

Step 5: the fixed point is unique. Suppose y^* is another fixed point. Then

$$\begin{aligned}\|x^* - y^*\| &= \|f(x^*) - f(y^*)\| \\ &\leq L\|x^* - y^*\|.\end{aligned}$$

Hence $(1 - L)\|x^* - y^*\| \leq 0$. Both factors on the left are nonnegative and $1 - L > 0$, so $\|x^* - y^*\| = 0$. Therefore $x^* = y^*$.

Step 6: error estimates. Letting $m \rightarrow \infty$ in the estimate from Step 2 gives

$$\|x_k - x^*\| \leq \frac{L^k}{1 - L} \|x_1 - x_0\|,$$

which is Eq. (1). For $k \geq 1$, applying the contraction inequality directly to x_{k-1} and x^* gives

$$\|x_k - x^*\| = \|f(x_{k-1}) - f(x^*)\| \leq L\|x_{k-1} - x^*\|.$$

Repeating this k times proves Eq. (2); for $k = 0$, that bound is equality.

For the a posteriori estimate, first sum the tail starting at x_k :

$$\begin{aligned}\|x_k - x^*\| &\leq \sum_{j=k}^{\infty} \|x_{j+1} - x_j\| \\ &\leq \frac{1}{1 - L} \|x_{k+1} - x_k\|.\end{aligned}$$

Since $\|x_{k+1} - x_k\| \leq L\|x_k - x_{k-1}\|$, we obtain Eq. (3). This completes all parts of the theorem. $\square$

4 Interpretation and necessity of the hypotheses

The proof explains why fixed-point iteration is useful algorithmically. The factor L^k gives geometric convergence, and Eq. (3) gives a stopping rule that does not require knowing x^* . If $L > 0$ and

$$\|x_k - x_{k-1}\| \leq \frac{1 - L}{L} \varepsilon,$$

then $\|x_k - x^*\| \leq \varepsilon$. When $L = 0$, the map is constant, and x_1 is already the fixed point.

Both major hypotheses have a purpose.

- The strict inequality $L < 1$ makes the geometric series converge and forces uniqueness. The map $f(x) = x + 1$ on $\mathbb{R}$ has Lipschitz constant 1 but has no fixed point.
- Completeness ensures that the Cauchy limit belongs to the domain. On the incomplete space $X = (0, 1)$, the map $f(x) = x/2$ is a contraction, but its only fixed point in $\mathbb{R}$ is $0 \notin X$.

The same proof works without coordinates: let (X, d) be any nonempty complete metric space, and let $f : X \rightarrow X$ satisfy $d(f(x), f(y)) \leq Ld(x, y)$ for all $x, y \in X$ and some $L \in [0, 1)$. Then f has a unique fixed point, and the iteration $x_{k+1} = f(x_k)$ converges to it from every $x_0 \in X$.